

SHETH MVLU COLLEGE

SUBJECT:- WSN

Aim:Wireless Sensor Network Deployment and Coverage Analysis

Simulate random and grid deployment of sensor nodes and evaluate network coverage and connectivity.

CODE:

```
set ns [new Simulator]

set tracefile [open wsn.tr w]
$ns trace-all $tracefile

set namfile [open wsn.nam w]
$ns namtrace-all-wireless $namfile 500 500

set topo [new Topography]
$topo load_flatgrid 500 500

set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802_11
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ant) Antenna/OmniAntenna
set val(ifqlen) 50
set val(nn) 9
set val(rp) AODV

create-god $val(nn)

set chan_1_ [new Channel/WirelessChannel]

$ns node-config \
    -adhocRouting $val(rp) \
    -llType $val(ll) \
    -macType $val(mac) \
    -ifqType $val(ifq) \
    -ifqLen $val(ifqlen) \
    -antType $val(ant) \
    -propType $val(prop) \
    -phyType $val(netif) \
    -channel $chan_1_ \
    -topoInstance $topo \
    -agentTrace ON \
    -routerTrace ON \
    -macTrace ON

for {set i 0} {$i < $val(nn)} {incr i} {
    set node_($i)[ns node]
}

$node_(0) set X_ 100
```

ANEES SAYED T111

SHETH MVLU COLLEGE

SUBJECT:- WSN

```
$node_(0) set Y_ 100

$node_(1) set X_ 200
$node_(1) set Y_ 100

$node_(2) set X_ 300
$node_(2) set Y_ 100

$node_(3) set X_ 100
$node_(3) set Y_ 200

$node_(4) set X_ 200
$node_(4) set Y_ 200

$node_(5) set X_ 300
$node_(5) set Y_ 200

$node_(6) set X_ 100
$node_(6) set Y_ 300

$node_(7) set X_ 200
$node_(7) set Y_ 300

$node_(8) set X_ 300
$node_(8) set Y_ 300

for {set i 0} {$i < $val(nn)} {incr i} {
    node_$i set Z_ 0
    $ns initial_node_pos node_$i 20
}

set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0

set null0 [new Agent/Null]
$ns attach-agent $node_(8) $null0

$ns connect $udp0 $null0

set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.1
$cbr0 attach-agent $udp0

nsat1.0 "cbr0 start"
nsat8.0 "cbr0 stop"

proc finish {} {
    global ns tracefile namfile
```

SHETH MVLU COLLEGE

SUBJECT:- WSN

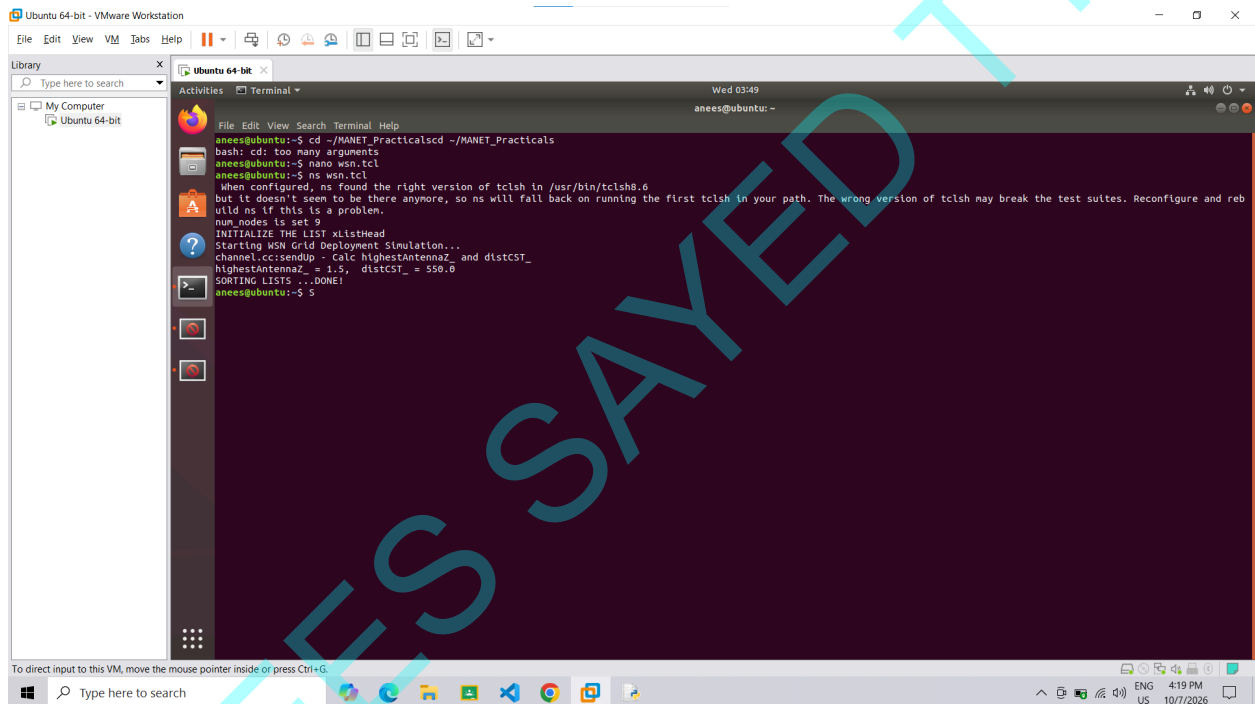
```
$ns flush-trace  
close $tracefile  
close $namfile
```

```
exec nam wsn.nam &  
exit 0  
}
```

\$ns at 10.0 "finish"

puts "Starting WSN Grid Deployment Simulation..."
\$ns run

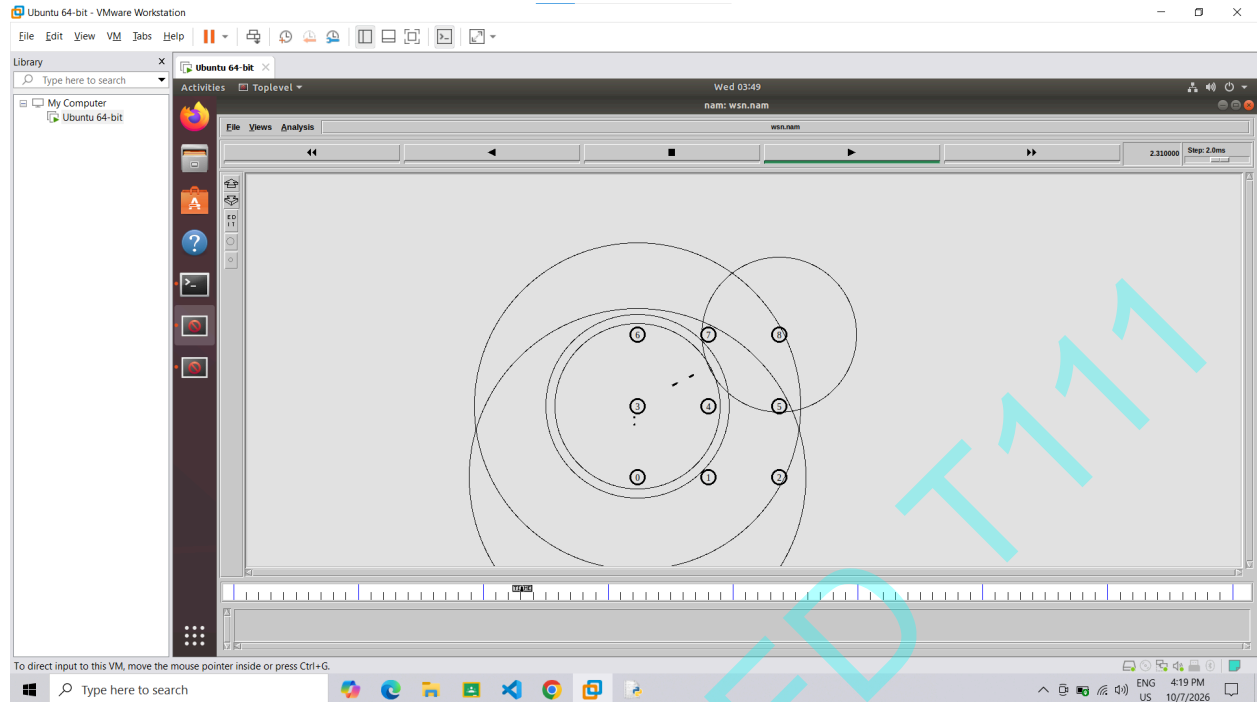
OUTPUT:



```
anees@ubuntu:~$ cd ~/MANET_Practicals  
anees@ubuntu:~/MANET_Practicals$ nano wsn.tcl  
anees@ubuntu:~/MANET_Practicals$ ns wsn.tcl  
When configured, ns found the right version of tclsh in /usr/bin/tclsh8.6  
but it doesn't seem to be there anymore, so ns will fall back on running the first tclsh in your path. The wrong version of tclsh may break the test suites. Reconfigure and reb  
uild ns if this is a problem.  
num_nodes is set 9  
INITIALIZE THE LIST xlistHead  
Starting WSN Grid Deployment Simulation...  
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_  
highestAntennaZ_ = 1.5, distCST_ = 550.0  
SORTING LISTS ...DONE!  
anees@ubuntu:~/MANET_Practicals$
```

SHETH MVLU COLLEGE

SUBJECT:- WSN



Observation

- Nine sensor nodes were deployed in a structured 3×3 grid pattern.
- Uniform spacing provided balanced network coverage.
- Node 0 acted as the source node and Node 8 as the destination.
- AODV successfully discovered routes between nodes.
- CBR traffic was transmitted without interruption.
- The grid topology maintained good connectivity among sensor nodes.
- Packet transmission was successfully visualized in NAM.

Result

The WSN grid deployment simulation was successfully executed. Sensor nodes were uniformly deployed, and data packets were transmitted successfully from the source node to the destination node.